

Week 2 Lab: Powering Your MCU

EE0827 Embedded System Applications

Overview

Build a regulated 3.3V power supply and learn proper decoupling techniques for MCU power pins. You will observe the effects of good and bad power delivery on MCU behavior, focusing on practical skills you'll use in every embedded project.

Time allocation: 90 minutes

- Power supply construction: 25 min
- Decoupling experiments: 25 min
- Code walkthrough: 20 min
- Documentation and cleanup: 20 min

Learning Objectives

By the end of this lab, you will be able to:

1. Build a working 3.3V regulated power supply
2. Explain why decoupling capacitors are needed near MCU power pins
3. Observe the effect of decoupling on power rail noise under GPIO load
4. Measure power rail quality with an oscilloscope
5. Read and understand bare-metal GPIO register code

Safety Gate

Safety Checklist (Complete Before Powering On)

- ☐ Bench supply current limit set to 200 mA BEFORE turning on
- ☐ All wiring verified against schematic before applying power
- ☐ Polarity checked on regulator (Vin, Vout, GND pins)
- ☐ Nucleo board disconnected from USB during external power experiments
- ☐ Oscilloscope probe ground connected to circuit ground only

Protecting Your Nucleo Board

1. **Never exceed 3.6V on VDD/VDDA** - most GPIOs are 5V-tolerant (FT) in digital mode, but power pins are not
2. **Never power from USB and external supply simultaneously** unless using proper isolation
3. **Always verify voltage before connecting** - measure first, connect second
4. **Current limit your supply** - protects against wiring mistakes

Required Parts and Tools

Hardware

- STM32 Nucleo-F401RE board
- Bench power supply (0-30V with current limiting)
- LM317T adjustable voltage regulator (TO-220)
- Resistors for voltage setting: $1 \times 2.2\text{k}\Omega$ (R1), $1 \times 3.3\text{k}\Omega$ (R2) — from kit
- Capacitors: $1 \times 100\text{nF}$ ceramic (regulator input), $1 \times 22\mu\text{F}$ ceramic (regulator output)
- Breadboard and jumper wires
- Multimeter
- Oscilloscope with probes

Software

- STM32CubeIDE (for GPIO stress test program)

! PRE-LAB SECTION

Complete the following BEFORE arriving at the lab session.

Pre-Lab Checklist

Complete before arriving at the lab:

- ☐ Read STM32F401 datasheet section on power supply (Section 6.1.6)
- ☐ Identify all VDD and VSS pins on the LQFP64 pinout diagram
- ☐ Locate the decoupling capacitors on the Nucleo board schematic
- ☐ Review LM317T pinout (ADJ, OUT, IN — TO-220 package)

Pre-Lab Questions

1. How many VDD pins does the STM32F401RE have? _____
2. What is the difference between VDD and VDDA? _____
3. What capacitor value is typically used for high-frequency decoupling? _____
4. Why must decoupling capacitors be placed close to the MCU pins? _____

i IN-LAB SECTION

The following sections are completed during the lab session.

Part 1

Build a 3.3V Power Supply

25 minutes

Step 1.1: LM317T Circuit

Build a regulated 3.3V supply on your breadboard using the LM317T adjustable regulator:

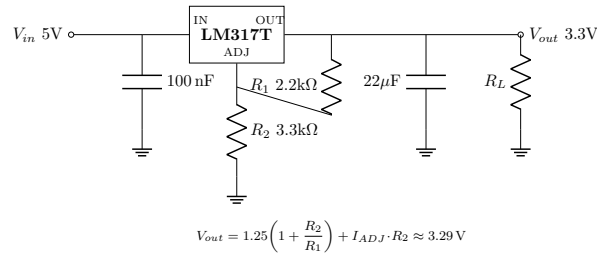


Figure 1: LM317T Regulator Circuit

LM317T pinout (TO-220, viewed from front):

- Pin 1: ADJ (adjustment)
- Pin 2: OUT (also metal tab)
- Pin 3: IN

Voltage setting resistors: The LM317T output voltage is set by two external resistors:

$$V_{out} = V_{ref} \left(1 + \frac{R_2}{R_1} \right) + I_{ADJ} \times R_2$$

where $V_{ref} = 1.25 \text{ V}$ and $I_{ADJ} \approx 50 \mu\text{A}$ (typical). With high-value resistors the I_{ADJ} term is **not** negligible.

With $R_1 = 2.2 \text{ k}\Omega$ (OUT $\rightarrow$ ADJ) and $R_2 = 3.3 \text{ k}\Omega$ (ADJ $\rightarrow$ GND):

$$V_{out} = 1.25 \times \left(1 + \frac{3.3 \text{ k}}{2.2 \text{ k}} \right) + 50 \mu\text{A} \times 3.3 \text{ k} = 3.125 + 0.165 = 3.29 \text{ V}$$

Step 1.2: Test Your Regulator

1. Set bench supply to **5V** (the LM317T needs $\sim 1.5\text{--}2\text{V}$ dropout headroom; 5V gives adequate margin)
2. Set current limit to **200 mA**
3. Connect to your circuit
4. Measure output voltage

Measurement	Expected	Your Value
Vout (no load)	$3.29\text{V} \pm 0.15\text{V}$	_____ V

Checkpoint 1: Regulator Working

Before proceeding to decoupling experiments, verify:

- ☐ Output voltage is $3.29\text{V} \pm 0.15\text{V}$
- ☐ No excessive heat from regulator
- ☐ Current limit still set to 200 mA

Why 5V Input?

Using 5V input instead of 12V means:

- Less heat in the regulator: $(5\text{V} - 3.3\text{V}) \times I$ vs $(12\text{V} - 3.3\text{V}) \times I$
- The Nucleo board's ST-Link provides 5V, so this matches real use
- The LM317T has a higher dropout ($\sim 1.5\text{--}2\text{V}$) than a fixed LDO ($\sim 1.1\text{V}$). With 5V in, you get $\sim 1.7\text{V}$ headroom — adequate at 100 mA but tighter than an AMS1117

For this lab, we care about clean 3.3V, not efficiency.

Part 2

Understanding Decoupling

20 minutes

This is the core of the lab. You will observe how decoupling capacitors affect MCU power quality and behavior.

Step 2.1: Examine Nucleo Board Power

The Nucleo board has proper decoupling built in. Let's find it.

Using the Nucleo schematic, locate:

1. The 3.3V LDO regulator (LD39015M33R)
2. Decoupling capacitors near the STM32F401 power pins
3. VDDA filtering components

Record what you find:

Component	Value	Location
VDD decoupling caps	_____	_____
VDDA filter caps	_____	_____
LDO output cap	_____	_____

Note

The Nucleo board does **not** include a bulk capacitor (4.7–10 μF) near the MCU pins, even though ST's own application notes recommend one. It relies on 100 nF caps per VDD pin only. Keep this in mind when designing your own boards.

Step 2.2: Measure Nucleo Power Quality

With the Nucleo powered via USB:

1. Connect oscilloscope probe to 3.3V rail (CN7 pin 16 or CN10 pin 7)
2. Connect probe ground to GND
3. Set oscilloscope:
 - Coupling: **AC** (to see only ripple/noise)
 - Scale: **20 mV/div**
 - Timebase: **10 μs /div**
 - Bandwidth limit: **20 MHz ON**

4. Observe the baseline noise

Measurement	Value
Noise (pk-pk) idle	_____ mV

Take screenshot: M_nucleo_idle.png

Step 2.3: Load the Power Rail

Flash the GPIO stress test program (`assets/code/gpio_stress_test.c`). It configures all Port A pins as outputs and toggles them at maximum speed, creating current transients on the 3.3V rail.

Note

This program reconfigures PA2/PA3 (UART), so you will lose serial output while it runs. Press the **RESET** button to restore normal operation afterward.

Observe the 3.3V rail on oscilloscope:

Measurement	Value
Noise (pk-pk) with GPIO activity	_____ mV

Take screenshot: M_nucleo_active.png

Why Does Decoupling Work?

The Nucleo's built-in 100 nF capacitors sit millimetres from the MCU power pins. When the GPIO logic switches, it draws sharp current pulses. The local capacitor supplies that charge instantly, so the voltage barely dips. Without it, the current must travel through long traces and wires whose inductance causes $V = L \times di/dt$ voltage drops — exactly the kind of noise that corrupts ADC readings and causes erratic behavior. That is why every MCU datasheet specifies a 100 nF capacitor on *each* VDD pin, placed as close as possible.

Part 3

Understanding the Stress Test Code

20 minutes

The GPIO stress test you flashed in Step 2.3 is a bare-metal C program — no HAL, no RTOS. Let's understand exactly what it does and why it affects the power rail.

Step 3.1: Code Walkthrough

```
#include "stm32f4xx.h"
#include <stdint.h>

#define TOGGLE_MASK 0xFFFF

int main(void) {
    /* Enable GPIOA clock */
    RCC->AHB1ENR |= RCC_AHB1ENR_GPIOAEN;

    /* Configure all Port A pins as output */
    GPIOA->MODER = 0x55555555; /* 01 per pin = output */

    /* High-speed slew rate for sharpest edges */
    GPIOA->OSPEEDR = 0xFFFFFFFF; /* 11 per pin = high */

    /* Toggle forever - no delay */
    while (1) {
        GPIOA->ODR ^= TOGGLE_MASK;
    }
}
```

Line-by-line explanation:

Line	What it does
RCC->AHB1ENR = ...GPIOAEN	Enables the clock to GPIO Port A. Without this, the GPIO peripheral is powered off and its registers cannot be accessed.
GPIOA->MODER = 0x55555555	Sets all 16 Port A pins to general-purpose output mode. Each pin uses 2 bits in MODER: 01 = output. Sixteen pins $\times$ 01 = 0101...0101 = 0x55555555.
GPIOA->OSPEEDR = 0xFFFFFFFF	Sets all pins to high-speed slew rate (~100 MHz). Each pin uses 2 bits: 11 = high speed. Faster edges mean sharper current transients on VDD.

Line	What it does
GPIOA->ODR ^= TOGGLE_MASK	XOR toggles all 16 output data register bits at once. Every pin that was HIGH goes LOW, and vice versa.
while (1) { ... }	Infinite loop with no delay — runs at maximum CPU speed.

Why Does This Create Power Rail Noise?

Each GPIO pin charges or discharges its parasitic capacitance ($\sim 5\text{pF}$) on every toggle. With 16 pins switching simultaneously at several MHz, the total *transient* current drawn from VDD can reach several mA in sharp spikes. The decoupling capacitors must supply this charge locally; any inductance in the path (long traces, wires) turns these current spikes into voltage dips ($V = L \times di/dt$).

Step 3.2: HAL Comparison

The same program written with the STM32 HAL library:

```
#include "main.h"

int main(void) {
    HAL_Init(); /* default 16 MHz HSI clock */

    __HAL_RCC_GPIOA_CLK_ENABLE();

    GPIO_InitTypeDef GPIO_InitStruct = {0};
    GPIO_InitStruct.Pin = GPIO_PIN_All;
    GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
    GPIO_InitStruct.Pull = GPIO_NOPULL;
    GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_VERY_HIGH;
    HAL_GPIO_Init(GPIOA, &GPIO_InitStruct);

    while (1) {
        HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_All);
    }
}
```

	Bare-metal	HAL
Clock enable	Direct register write	<code>__HAL_RCC_GPIOA_CLK_ENABLE()</code> macro

	Bare-metal	HAL
Pin config	2 register writes (MODER + OSPEEDR)	HAL_GPIO_Init() — sets MODER, OTYPER, OSPEEDR, PUPDR in one call
Toggle	ODR ^= mask (1 instruction)	Function call → read ODR → XOR → write BSRR
Loop overhead	~6 cycles	~15–20 cycles

Both versions set high-speed slew rate. The HAL version toggles **~3× slower** due to function call overhead in the loop, making the bare-metal version more effective as a stress test.

Step 3.3: Register Questions

1. The MODER register uses 2 bits per pin. What are the four possible modes? (Hint: check the STM32F401 reference manual, Section 8.4.1.) _____
2. Why does 0x55555555 set all pins to output? Write out the first 8 bits in binary and identify which pins they control. _____
3. At 16 MHz (default HSI, no PLL), the read-modify-write + branch loop takes ~6 CPU cycles. How many toggles per second does this produce? _____

Measurement Summary

ID	Measurement	Expected	Measured	Pass
M1	Regulator output voltage	$3.29\text{V} \pm 0.15\text{V}$		
M2	Nucleo 3.3V noise (idle)	$< 20\text{ mV pk-pk}$		
M3	Nucleo 3.3V noise (GPIO active)	$< 50\text{ mV pk-pk}$		

Failure Injection Test

Demonstrate understanding by predicting failure modes:

Scenario	What Happens?	How to Detect/Prevent?
Input/output capacitors removed		
Regulator Vin and Vout swapped		
Decoupling cap far from MCU (long wires)		
VDDA connected to unfiltered VDD		
Ground connection missing/loose		

Analysis Questions

Answer these during the lab (2-3 sentences each):

1. Why do decoupling capacitors need to be placed close to the MCU power pins?

2. What would happen if you powered an MCU with no decoupling capacitors at all? _____

3. Why does the STM32 have separate VDDA and VDD supplies? _____

4. If the VDDA supply had 50 mV of noise, how would that affect 12-bit ADC readings? (Hint: what is the voltage per LSB at 3.3 V reference?) _____

Completion Checklist

Before leaving the lab, verify you have accomplished:

- ☐ Working 3.3V regulator (verified output voltage)
- ☐ Screenshot: Nucleo power rail noise (idle and active)
- ☐ Completed measurement summary table
- ☐ Answered analysis and register questions

Success Criteria

This lab is ungraded practice. Use these criteria to gauge your progress:

Area	What to Aim For
Regulator construction	Working 3.3V output with correct capacitors
Decoupling experiments	Noise difference between idle and GPIO-active Nucleo
Code understanding	Correct answers to MODER, binary, and toggle-rate questions
Analysis questions	Thoughtful answers demonstrating understanding

Common Issues and Solutions

Issue	Symptom	Solution
No regulator output	0V	Check polarity, verify input connected
Output = Input	No regulation	Vout and Vin pins swapped; check R1/R2 and ADJ pin connection
Output voltage wrong	Not ~3.29V	Verify R1 (2.2k Ω) between OUT-ADJ and R2 (3.3k Ω) between ADJ-GND
Supply trips at power-on	Voltage collapses	Capacitor inrush current; raise current limit to 200 mA
Can't see ripple	Flat line	AC coupling on, reduce V/div scale
Excessive noise	> 100mV	Use short scope ground lead
Nucleo won't start	No activity	Verify 3.3V present at VDD pins

Cleanup Checklist

Before leaving:

- ☐ Disconnect all external power from Nucleo
- ☐ Turn off bench power supply
- ☐ Return oscilloscope probes
- ☐ Save all screenshots and data
- ☐ Clean up breadboard area

Key Takeaways

1. **Every MCU needs decoupling capacitors** - typically 100nF per VDD pin plus a bulk cap
2. **Placement matters** - capacitors must be close to the pins they're decoupling
3. **VDDA needs extra filtering** - for clean ADC readings
4. **Measure your power rails** - oscilloscope in AC coupling mode shows what your multimeter misses
5. **The Nucleo board does this right** - study its schematic as a reference design